



NORTH SYDNEY BOYS HIGH SCHOOL

2010 YEAR 12 HSC ASSESSMENT TASK 1

Mathematics

General Instructions

- Working time – 50 minutes
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

Total Marks (50)

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Fletcher
- ☐ Mr Rezcallah
- ☐ Mr Lowe
- ☐ Mr Ireland
- ☐ Mr Barrett
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number:

Question	1	2	3	4	5	Total	Total
Mark	9	13	8	11	9	50	100

Question 1 (9 marks) *Start a new page*

Marks

(a) Differentiate with respect to x :

(i) $\frac{x^2+2x}{x}$ 2

(i) $(4 - x^2)^3$ 2

(iii) $\frac{3x+1}{2x-1}$ 2

(b) Find the equation of the tangent to the curve $y = x^3 - 2x + 2$ at the point $(2, 6)$ 3

Question 2 (13 marks) *Start a new page*

(a) Evaluate $\sum_{n=1}^5 n^2$ 1

(b) The third term of an arithmetic progression is 21 and the 9th term is 57.

(i) Find the value of the first term and the common difference. 2

(ii) Find the sum of the first 80 terms. 2

(c) Find the 10th term of the geometric series $-5, 10, -20, \dots$ 3

(d) Frank keeps a big stack of paint cans in his hardware store. The stack has 3 cans on the top row and each succeeding row has 2 more cans than the previous row. (A diagram may be helpful here.)

(i) Find the number of cans in the 12th row. 2

(ii) If there are 224 cans in the stack, how many rows are there? 3

Question 3 (8 marks) *Start a new page*

(a) Consider the graph of $y = x^3 + 3x^2 - 9x - 11$

(i) Show that this function has stationary points at $(1, -16)$ and $(-3, 16)$ and determine their nature. 3

(ii) Find any points of inflexion. 2

(iii) Sketch the curve, showing all important features (not x -intercepts) 2

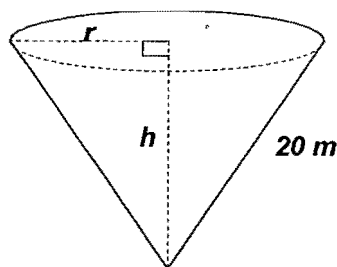
(iv) For what values of x is the curve increasing? 1

Question 4 (11 marks) *Start a new page*

(a) A particle is moving in a straight line so that its distance, x metres, from a fixed point, O , after t seconds is given by $x = \frac{1}{3}t^3 - \frac{7}{2}t^2 + 6t$.

- (i) Find the equation for its velocity. 1
- (ii) Find its initial position and velocity. 2
- (ii) When is the particle at rest? 2

(b) A cone shaped storage tank is constructed. It has a slant height of 20 metres. The top has radius r metres and the whole tank is h metres high, as shown in the diagram:



- (i) Express r in terms of h . 1
- (i) Show that the volume, $V \text{ m}^3$ of the tank can be expressed by
$$V = \frac{\pi}{3}(400h - h^3)$$
 2
- (ii) Hence find the value of h which will give the cone its maximum volume. 3

Question 5 (9 marks) *Start a new page*

(a) “The inflation level I is still increasing, but at a reduced rate of increase.”

- (i) Sketch a graph to illustrate the above statement. 1
- (ii) What does the above statement mean with reference to

$$\frac{dI}{dt} \text{ and } \frac{d^2I}{dt^2} ? \quad \text{2}$$

(b) A container holds 50 litres of water. A pump extracts 10 litres on the first cycle and 7.5 litres on the second. In each future pumping cycle the pump extracts $\frac{3}{4}$ of the amount that the previous cycle extracted.

Show that the container will never be emptied, and find how much water will finally remain in the container. 3

(c) If the sum of the first n terms of a sequence is given by

$$S_n = n(2n - 1), \text{ find a formula for the } n^{\text{th}} \text{ term of the sequence, } T_n. \quad \text{3}$$

Q1 (a) (i) $\frac{d}{dx} \left(\frac{x^2+2x}{x} \right) = \frac{d}{dx} (x+2)$ ✓
 $= 1$ ✓

(ii) $\frac{d}{dx} \left((4-x^2)^3 \right) = 3 \cdot (4-x^2)^2 \cdot -2x$ ✓
 $= -6x (4-x^2)^2$ ✓

(iii) $\frac{d}{dx} \left(\frac{3x+1}{2x-1} \right) = \frac{(2x-1)(3) - (3x+1)(2)}{(2x-1)^2}$ ✓
 $= \frac{6x-3-6x-2}{(2x-1)^2}$
 $= \frac{-5}{(2x-1)^2}$ ✓

(b) $y = x^3 - 2x + 2$

$y' = 3x^2 - 2$ ✓

$\therefore$ at $(2, 6)$, $y' = 3 \cdot 2^2 - 2$ ✓
 $= 10$

$\therefore y - 6 = 10(x - 2)$ ✓

$y = 10x - 14.$

Q2 (a) $\sum_{n=1}^5 n^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$
 $= 545$ ✓

(b) (i) $T_3 = a + 2d = 21$
 $T_9 = a + 8d = 57$

$\therefore 6d = 36$

$d = 6$ ✓

$a = 9$ ✓

(ii) $S_{80} = \frac{80}{2} (2 \times 9 + 79 \times 6)$ ✓
 $= 19\,680$ ✓

(c) $r = \frac{10}{-5} = -2$ ✓

$\therefore T_{10} = -5 \cdot (-2)^9$ ✓
 $= 2560$ ✓

(d) (i) $a = 3, d = 2$
 $T_{12} = 3 + 11 \times 2$ ✓
 $= 25$ ✓

(ii) $224 = \frac{n}{2} (6 + 2(n-1))$ ✓
 $= n(3 + n - 1)$
 $= n(n+2)$

$\therefore n^2 + 2n - 224 = 0$ ✓

$(n+16)(n-14) = 0$

$\therefore n = 14$ (as $n > 0$)

ie. 14 rows ✓

Q3 (a) $y = x^3 + 3x^2 - 9x - 11$

(i) $y' = 3x^2 + 6x - 9$
 $= 3(x+3)(x-1)$ ✓

$y'' = 6x + 6$
 $= 6(x+1)$

For stat. point, $y' = 0 \therefore x = -3$ or $x = 1$ } ✓
 $y = 16$ $y = -16$

At $(-3, 16)$, $y'' = -12 < 0 \therefore$ max. turn. pt. at $(-3, 16)$ } ✓

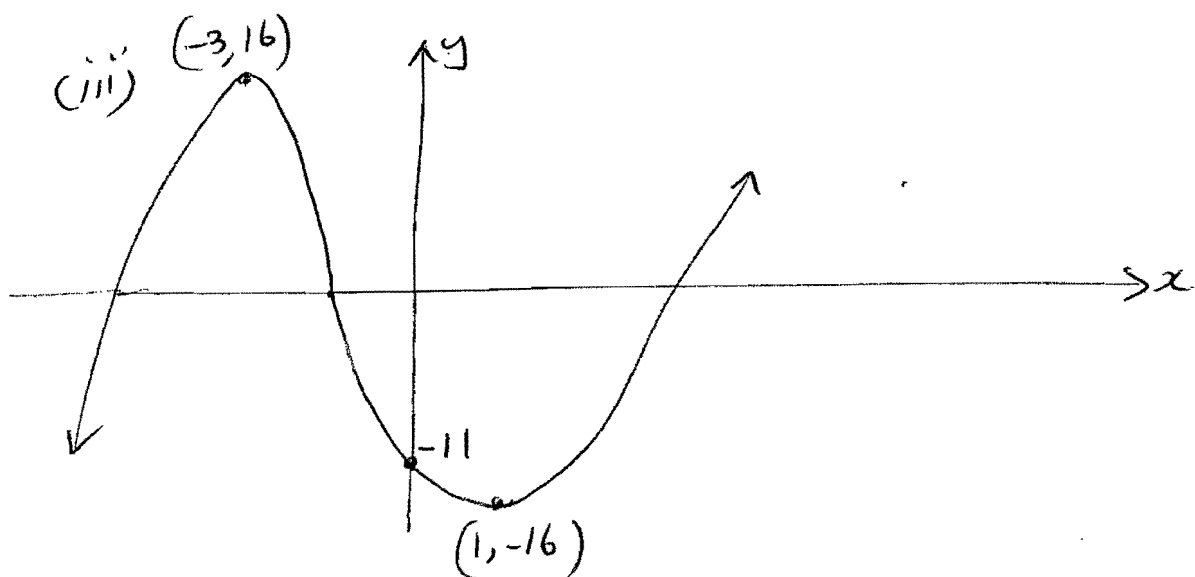
At $(1, -16)$, $y'' = 6(2) > 0 \therefore$ min. turn. pt. at $(1, -16)$ }

(ii) For inflexions, $y'' = 0 \therefore x = -1$ ✓
 $y = 0$

Test :-

x	-2	-1	0
y''	-	0	+

 $\therefore$ change of concavity
 $\therefore$ inflexion at $(-1, 0)$. ✓



(iv) From the graph, y is increasing
for $x < -3$ or $x > 1$ ✓

Q4 (a) $x = \frac{1}{3}t^3 - \frac{7}{2}t^2 + 6t$

(i) $v = \frac{dx}{dt} = t^2 - 7t + 6$ ✓

(ii) at $t=0$, $x=0$ $\therefore$ at origin ✓

$v = 6$ $\therefore$ 6m/s to right. ✓

(iii) "at rest" $\Rightarrow v=0$

$\therefore (t-1)(t-6) = 0$ ✓

$\therefore t=1, 6$

So $\therefore$ at rest at 1 second & at 6 seconds. ✓

(b) (i) $r^2 = 20^2 - h^2$

$\therefore r = \sqrt{400 - h^2}$ ✓

(ii) $V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$

$= \frac{1}{3} \pi (400 - h^2) h$

$V = \frac{\pi}{3} (400h - h^3)$ ✓✓

(iii) For max., $\frac{dV}{dh} = 0$

$\therefore \frac{\pi}{3} (400 - 3h^2) = 0$ ✓

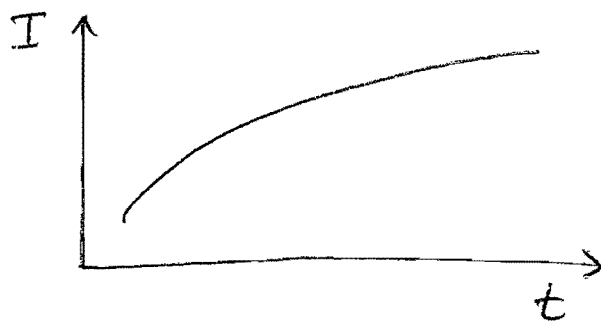
$3h^2 = 400$

$h = \frac{20}{\sqrt{3}}$

(as $h > 0$) ✓

Test: $\frac{d^2V}{dh^2} = \frac{\pi}{3} (-6h) < 0$

$\therefore$ max. at $h = \frac{20}{\sqrt{3}}$ m. ✓

Q5 (a) (i)

(ii) $\frac{dI}{dt} > 0$ means inflation is increasing. ✓

$\frac{d^2I}{dt^2} < 0$ means the rate of increase is slowing down. ✓

(b) The amounts extracted form an infinite G.P. with $a=10$, $r=\frac{3}{4}$, ✓

Since water extracted = $10 + 7.5 + \dots$

$$\therefore S_{\infty} = \frac{10}{1 - \frac{3}{4}}$$

$$= 40 \text{ litres}$$

$\therefore$ 10 litres will remain. ✓

(c) $T_n = S_n - S_{n-1}$ ✓

$$= n(2n-1) - (n-1)(2(n-1)-1)$$

$$= 2n^2 - n - (n-1)(2n-3)$$

$$= 2n^2 - n - (2n^2 - 5n + 3)$$

$$= 4n - 3$$

$$\therefore T_n = 4n - 3$$
 ✓